Task 0.1:

$$I = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & -3 & -4 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix} \qquad K = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

We need to pad the "image" in order to achive same Convolution.

$$I_{padded} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 2 & 1 & 0 \\ 0 & 1 & -3 & -4 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Note: The green box represents the sliding Window (Kernal).

Answer:

$$O = \begin{bmatrix} 4 & 5 & 6 & 4 \\ 5 & 3 & 3 & 6 \\ 1 & -7 & -7 & 0 \\ 4 & 1 & 0 & 4 \end{bmatrix}$$

Task 0.2:

Apply Relu to the output of the last step:

$$A_{Relu} = \begin{bmatrix} 4 & 5 & 6 & 4 \\ 5 & 3 & 3 & 6 \\ 1 & 0 & 0 & 0 \\ 4 & 1 & 0 & 4 \end{bmatrix}$$

Task 0,3 :

$$p_{max} = \begin{bmatrix} 5 & 6 \\ 4 & 4 \end{bmatrix}$$

Task 0,4 :

$$X = [5 \quad 6 \quad 4 \quad 4]$$

Task 0,5 :

Let the FC layer contain 2 neurons.

Let W_I and W_{II} be the weights of each neuron.

Where: $W_I = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$, $W_{II} = \begin{bmatrix} 2 \\ 2 \\ 2 \\ 2 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

Then the weight matrix : $W = \begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 1 & 2 \\ 1 & 2 \end{bmatrix}$

Answer :

$$y = xW + b = \begin{bmatrix} 19 \\ 38 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 20 \\ 40 \end{bmatrix}$$

Task 0,6 :

$$\sigma(y)_I = \frac{e^{20}}{e^{20} + e^{40}} = 2 \cdot 10^{-9}$$

$$\sigma(y)_{II} = \frac{e^{40}}{e^{20} + e^{40}} = 0{,}9999$$

Task 0,7 :

$$H(y, g) = -(\log(0{,}6)) = 0{,}2218$$

$$MSE(y, g) = \frac{1}{3}\left((0{,}25 - 0)^2 + (0{,}6 - 1)^2 + (0{,}15 - 0)^2\right) = 0{,}0816$$

$$SVM(y, z) = \left(\max(0; 0{,}25 - 0{,}6 + 1) + \max(0; 0{,}15 - 0{,}6 + 1)\right) = 1{,}2$$

Evaluation metrics

Task 0,1 :

1. In situations where precision and recall are important, commonly in medical applications

2. The problem with accuracy is that it treats every class equally regardless of the class frequency or spatial overlap like the jaccard index do.

3. F1 is the harmonic mean over the dataset, while jaccard index measures similarity between sample sets and is defined as the size of intersection

divided by the union. Think Venndiagram.

Both are good in those area of application.

F1-score is good on unbalanced data and where balance between precision and recall is needed

Jaccard Index is good on spatial accuracy. it is a standard metric on object detection algorithms.

Task 0.2:

class	GT	P	Correct (∩)	Total (U)
B	1, 3, 4	1, 2, 4, 5	1, 4	1, 2, 3, 4, 5
T	2, 5, 6, 7, 8	3, 6, 7, 8	6, 7, 8	2, 3, 5, 6, 7, 8
D	5, 6, 7, 8	3, 4, 5, 8	5, 8	3, 4, 5, 6, 7, 8
C	0	5, 6	0	5, 6

class	Fequency	Jaccard	precision	Recall	F1
B	3/12	2/5	2/4	2/3	4/7
T	5/12	3/6	3/4	3/5	6/9
D	4/12	1/6	1/3	1/4	2/7
C	0	0/2	0/2	0	0

$$\text{mean Jaccard} : \frac{\frac{2}{5} + \frac{3}{6} + \frac{1}{6}}{4} = 0.267$$

mean F_1 score : $\dfrac{\frac{4}{7} + \frac{6}{9} + \frac{2}{7}}{4} = 0,381$

Frequency mean Jaccard : $\dfrac{\frac{3 \cdot 2}{5} + \frac{5 \cdot 3}{6} + \frac{4 \cdot 1}{6}}{12} = 0,364$

Frequency mean F_1 score : $\dfrac{\frac{3 \cdot 4}{7} + \frac{5 \cdot 6}{9} + \frac{4 \cdot 2}{7}}{12} = 0,516$